Agentic AI System for Type 2 Diabetes Monitoring and Management

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Finally, I would like to acknowledge the importance of the data and clinical guidelines that made this project possible. This work is dedicated to the advancement of healthcare technology, with the hope that agentic AI can one day improve the quality of life for those living with Type 2 Diabetes.

Abstract

Type 2 Diabetes management suffers from data fragmentation and delayed clinical intervention due to a reliance on periodic manual monitoring. Current healthcare models lack the capability to provide real-time, actionable insights from patient-generated data. This creates a critical need for an automated, intelligent system that bridges the gap between continuous IoT data collection and professional medical oversight.

The proposed solution is an Agentic AI system that integrates real-time IoT monitoring with a clinical decision support framework to automate the diabetes treatment journey. This design provides a multi-layered approach to patient care, combining predictive risk assessment with generative AI for patient education. The system ensures safety through a "Doctor-in-the-Loop" architecture, where all AI-generated clinical suggestions require professional validation before implementation.

The development process follows a hybrid architectural approach. A calibrated Logistic Regression model is implemented using Scikit-learn to categorize patient risk into Low, Medium, or High categories based on physiological features such as BMI, glucose levels, and blood pressure. A rule-based engine generates medication adjustments, such as Metformin or Insulin titration, based on clinical HbA1c thresholds. For patient interaction, a Llama-3.2-1B-Instruct model is fine-tuned via the Unsloth library on a 5,000-record diabetes dataset. This ensures the model provides accurate, context-aware explanations of model coefficients and medical symptoms. Data management is handled through a dual-database architecture: MySQL is utilized for structured relational data and user management, while PostgreSQL with the TimescaleDB extension is employed to optimize the storage and analysis of high-frequency time-series data from IoT devices. All services are orchestrated via a FastAPI backend.

The system is expected to reduce the time between physiological changes and treatment adjustments while increasing patient health literacy through AI-driven explanations. Implementation results indicate high precision in risk stratification and reliable clinical suggestion generation. Future research should focus on integrating wearable sensor APIs for direct live-data ingestion and expanding the model to support multi-morbidity chronic care.

Keywords: Agentic AI, Diabetes Management, IoT Healthcare, Time-series Data, Machine Learning.

Introduction

Problem Description and Motivation

The Global Burden of Diabetes Diabetes Mellitus, specifically Type 2 (T2DM), has evolved into one of the most significant global health crises of the 21st century. According to the International Diabetes Federation (IDF), hundreds of millions of people live with this condition, a number projected to rise significantly by 2045. The management of T2DM is not a periodic medical task but a continuous, 24/7 lifestyle requirement. From a system engineering perspective, Type 2 Diabetes management can be formally modeled as a continuous control system. In classical control theory, a system is defined by inputs, internal states, outputs, and feedback mechanisms. In the context of diabetes, physiological variables such as blood glucose concentration, insulin sensitivity, blood pressure, and metabolic rate represent the system states. External inputs include dietary intake, physical activity, medication dosage, and behavioral factors, while clinical outcomes such as glycemic stability and cardiovascular health represent system outputs. In traditional healthcare models, this control system operates with delayed feedback loops. Clinical measurements are taken intermittently, and control actions (medication changes or lifestyle recommendations) are applied long after the physiological deviation has occurred. This delayed feedback creates system instability and increases the probability of oscillatory behavior, where patients alternate between hyperglycemia and hypoglycemia cycles.The absence of real-time feedback violates fundamental principles of control system stability. In engineering terms, the system lacks continuous sensing, real-time state estimation, and predictive control. This motivates the need for a closed-loop intelligent control architecture where sensing, analysis, decision-making, and intervention occur continuously rather than episodically. Patients must balance caloric intake, physical activity, and complex medication regimens while monitoring blood glucose levels.

The Monitoring Gap In the current clinical paradigm, patients typically visit a physician every 3 to 6 months. Between these visits, a "data black hole" exists. While patients may use home-testing kits, the data is often recorded manually, leading to human error or loss of information.

Furthermore, patients often lack the medical literacy to interpret sudden fluctuations in their data, leading to either unnecessary anxiety or dangerous negligence of symptoms.

Motivation for an Agentic Solution The motivation for this project stems from the potential of Agentic AI—artificial intelligence that does not just respond to prompts but acts as a proactive agent. By integrating Internet of Things (IoT) devices with advanced Machine Learning (ML) and Large Language Models (LLMs), we can create a "Digital Twin" for the patient. This system provides:

- Proactive Safety: Detecting risks before they become emergencies.
- Physician Empowerment: Providing doctors with organized, prioritized data and suggestions rather than raw, unanalyzed logs.
- Patient Agency: Giving patients the tools to understand their own health through natural language, reducing the psychological burden of chronic disease management.

Problem Statement and Objectives

Problem Statement

Despite the availability of glucose monitors and health tracking software, there is a lack of integrated systems that combine real-time time-series data analysis, predictive risk stratification, and automated clinical decision support with a human-in-the-loop safety mechanism. Existing solutions are either purely administrative or lack the intelligent reasoning required to suggest medication adjustments that adhere to clinical guidelines (like the ADA or AACE standards).

Risk Propagation and Objectives

Chronic diseases such as Type 2 Diabetes exhibit risk propagation behavior, where small physiological deviations can cascade into large-scale clinical complications over time. For example, sustained hyperglycemia can progressively lead to nephropathy, neuropathy, and cardiovascular disease. This project models risk as a dynamic propagating variable rather than a static condition. Risk accumulation is influenced by temporal patterns, lifestyle behaviors, and treatment adherence. Without continuous monitoring and early correction, risk states propagate across physiological subsystems. This justifies the need for continuous risk modeling and dynamic risk stratification, where patient states are re-evaluated continuously using time-series analysis and predictive models.

Project Objectives:

To solve this, the project focuses on the following primary objectives:

- Real-Time Data Ingestion: Design an architecture capable of receiving and storing continuous streams of IoT data (glucose, blood pressure, bmi, hba1c, lifestyle) using specialized time-series databases.
- Risk Stratification Engine: Develop a calibrated Logistic Regression model to classify patient states into Low, Medium, or High-risk categories. This engine is further designed to explain clinical values to the user by parsing model coefficients and feature thresholds to provide narrative health insights.
- Clinical Decision Support System (CDSS): Create a rule-based agent that generates therapeutic suggestions (e.g., Metformin, GLP-1 RA, or Insulin initiation) based on a structured clinical logic table derived from official guidelines (ADA, AACE, WHO). This agent evaluates patient metrics such as HbA1c, BMI, and SBP to suggest precise medications and dosages for physician review.
- Doctor-in-the-Loop Framework: Implement a secure interface for medical professionals to review, modify, or approve AI-generated suggestions.
- Fine-tuned Conversational Agent: Train and deploy a localized LLM (Llama 3.2-1B) to provide explanations of health metrics and answer patient queries using a curated diabetes knowledge base.
- Hybrid Data Management: Engineering a dual-database system utilizing MySQL for relational data and PostgreSQL/TimescaleDB for time-series IoT data storage to ensure system scalability and performance.
- Secure Telemedicine Communication Channel: To develop an integrated, real-time chat interface that facilitates direct communication between the doctor and the patient. This module allows the doctor to explain AI-suggested medication changes and enables the patient to seek clarification on their personalized risk reports, ensuring that the technology supports—rather than replaces—the human clinical relationship.

Closed-Loop Clinical Feedback Architecture

The proposed system implements a closed-loop feedback architecture. Sensor data forms the input layer, AI models perform state estimation and risk analysis, clinical logic engines generate control actions, and doctors validate interventions before execution. This feedback loop ensures continuous system regulation. Each clinical action modifies patient state, which is then re-measured and re-evaluated by the system. This creates a stable regulation cycle that aligns with engineering feedback control principles. The Doctor-in-the-Loop framework ensures that the feedback loop remains human-supervised, preventing unsafe autonomous behavior while preserving the benefits of intelligent automation.

Software and Hardware Requirements

The development of an Agentic AI system for Type 2 Diabetes management requires a sophisticated multi-layered stack. The following requirements ensure high availability for IoT data ingestion, low-latency AI inference, and secure, real-time communication between patients and physicians.

Software Requirements

Large Language Model (LLM) and Generative AI:

- Llama-3.2-1B-Instruct: The core Large Language Model used as the agentic conversational engine. It provides the reasoning capabilities required to answer patient queries and explain clinical metrics.
- Unsloth: A specialized library used for the memory-efficient fine-tuning of the Llama model. It is essential for optimizing the 1B-parameter model on the 5,000-record diabetes Q&A dataset.
- Hugging Face Transformers & PEFT: Frameworks used for model loading and Parameter-Efficient Fine-Tuning (LoRA/QLoRA), allowing the model to be specialized without requiring massive computational resources.

Backend and Orchestration Layer:

- Python 3.10+: The primary programming language for the entire ecosystem.

- FastAPI: A high-performance web framework used for building the system's APIs and handling asynchronous tasks.
- WebSockets: Integrated for the real-time, bidirectional chat interface between the doctor and the patient.
- Pydantic: Used for strict data validation of IoT payloads and clinical inputs.
- CryptContext (PassLib): A critical security component used for password hashing and sensitive data encryption. This ensures that patient and doctor credentials are stored securely in the MySQL database using industry-standard algorithms (like BCrypt), addressing the privacy constraints of medical software.
- JSON (JavaScript Object Notation): The primary data interchange format used throughout the system. It enables the seamless transmission of complex medical telemetry between IoT sensors, the FastAPI backend, and the frontend dashboards.

Machine Learning, Data Processing, and Risk Analytics:

- Pandas & NumPy: Essential for cleaning the 9,695 NHANES records and performing vector transformations on feature arrays.
- Scikit-Learn: The framework used to build the Logistic Regression classifier. It facilitates the construction of a ColumnTransformer pipeline for scaling numerical features and one-hot encoding categorical variables.
- CalibratedClassifierCV: A critical component from Scikit-learn used to apply Platt Scaling (Sigmoid method) to the model outputs. This ensures that the risk scores represent true probabilities, which are then mapped to Low (<0.35), Medium (0.35-0.7), and High (>0.7) risk categories.
- Matplotlib & Seaborn: Used for Exploratory Data Analysis (EDA) and visualizing the Confusion Matrix to evaluate model performance.

Database Architecture (Hybrid System):

MySQL (Relational Layer) Used for structured, non-temporal data that defines the system's users and static configurations:

- Doctors: Stores professional credentials, hashed passwords ($argon2id$...), and verification status.
- Patients: Manages user demographics (Gender, Ethnicity, DOB) and links each patient to an assigned doctor's email.
- Devices: Tracks IoT hardware inventory, including serial numbers, manufacturers (e.g., Abbott), and models.
- Suggestions: Logs system and doctor suggestions, including the medication category, dosage details in JSON format, and recommendation status (proposed/active/stopped/rejected).

PostgreSQL + TimescaleDB (Time-Series Layer) Used for time-stamped clinical data (ts) where identifying trends is critical for the AI models:

- Measurements: Ingests continuous telemetry for HbA1c, Glucose, and Blood Pressure, including values and units.
- Medications: Records the history of drug administration (e.g., Metformin) and dosage changes to track adherence over time.
- Lifestyle logs: Captures temporal behavioral data such as activity minutes and sleep hours.
- Predictions: Stores the historical output of the Risk Engine, including the probability score, risk category, and clinical explanation.

Hardware Requirement

The Implantable Sensing Mechanism:

The system centers on a miniaturized, pill-sized sensor (approximately 3.5 mm x 18 mm) that is medically implanted in the subcutaneous space of the upper arm. This hardware design is simulated; actual implantation is outside project scope.

- Continuous Glucose Monitoring (CGM): The sensor uses a unique fluorescence-based polymer that reacts to glucose levels in the interstitial fluid. It provides a real-time stream of readings every 5 minutes.
- HbA1c Estimation: While HbA1c is traditionally a lab-based blood test, the edge layer calculates an Estimated A1c (eA1c) or Glucose Management Indicator (GMI) by

aggregating the continuous glucose data over a 90-day window. This allows the doctor to see long-term trends without the patient needing frequent venipuncture.

Data Bridge: On-Body Transmitter:

Because the sensor is planted under the skin, it is an electrically passive device that does not contain a battery.

- Inductive Powering: A smart transmitter is worn on the surface of the skin directly over the implant. It uses Near-Field Communication (NFC) or inductive coupling to power the sensor and "interrogate" it for data.
- Processing & Vibration Alerts: The transmitter serves as the first "Edge" node. It processes the raw signal, converts it into a glucose value, and can provide discreet on-body vibratory alerts even if the patient's smartphone is not nearby.
- Wireless Uplink: The transmitter then sends the validated data via Bluetooth Low Energy (BLE) to the gateway (Smartphone/ESP32) for storage in the PostgreSQL/TimescaleDB backend.

Project Outcome

The primary outcome of this project is a fully integrated Agentic AI Healthcare Platform designed for the proactive management of Type 2 Diabetes. The system transitions from raw data to clinical action through four key deliverables:

Preventive Medicine as an Engineering Objective

This project reframes preventive medicine as an engineering objective rather than a purely clinical concept. By applying predictive modeling, continuous monitoring, and agentic reasoning, the system shifts healthcare from corrective intervention to preventive regulation. Engineering preventive healthcare systems requires early anomaly detection, predictive risk modeling, and intervention prioritization. The platform achieves this by integrating calibrated risk prediction, rule-based clinical logic, and real-time data ingestion. This transforms diabetes management into a preventive cyber-physical system where health deterioration is intercepted before irreversible damage occurs.

Interpretable Risk Stratification

The system delivers a calibrated Logistic Regression engine that classifies patients into Low, Medium, or High risk categories. Unlike "black-box" models, it provides narrative explanations for its predictions, such as identifying kidney stress when the ACR (Albumin-to-Creatinine Ratio) is elevated or flagging high HbA1c levels as a primary risk driver.

Automated Clinical Decision Support (CDSS)

The CDSS acts as the system's "Reasoning Agent," transforming raw physiological data from the TimescaleDB measurements table into actionable medical interventions. By mapping real-time values such as HbA1c, Glucose, and SBP against a deterministic logic matrix derived from clinical guidelines, the system identifies therapeutic gaps and generates precise suggestions. These outputs are stored in the MySQL suggestions table with a defined priority level and a JSON-formatted clinical rationale, covering actions such as initiating Metformin monotherapy, escalating to Triple therapy, or transitioning to Insulin initiation. This architectural design ensures a "Doctor-in-the-Loop" workflow, where AI-generated titration proposals remain in a "proposed" state until his doctor validates the treatment adjustment through the secure dashboard.

Agentic Conversational Assistant

The project results in a fine-tuned Llama-3.2-1B model specialized in diabetic education. This agent provides patients with 24/7 access to medical literacy, explaining complex lab results (like HDL or Serum Creatinine) in simple, natural language while maintaining a "Doctor-in-the-loop" safety protocol.

Real-Time Physician Dashboard & Monitoring

The final deliverable includes a secure web interface where physicians can:

- Monitor real-time IoT telemetry (Glucose, BP) stored in the TimescaleDB time-series backend.
- Review, accept, or reject AI-generated medication suggestions stored in the MySQL database.
- Engage in bidirectional chat with patients to validate treatment adjustments and lifestyle changes.

Constraints & Limitations

Developing a medical AI system involves strict boundaries regarding data privacy, technical performance, and clinical safety. These constraints shaped the architecture of the platform.

Technical & Connectivity Constraints

- Latency in Critical Alerts: While the PostgreSQL/TimescaleDB backend is optimized for speed, real-time alerts depend on the patient's internet stability. The system is constrained by "Network Jitter," meaning a delay in the cloud uplink could delay a high-priority notification to the doctor.
- Regulatory and Professional Boundaries: A primary constraint of this project is the legal and ethical boundary regarding medical procedures. As the lead engineer is not a licensed medical professional, the project is strictly prohibited from performing the physical implantation of subcutaneous sensors in human subjects. This constraint restricts the current scope to a "Hardware-in-the-Loop" simulation, where the software processes real datasets (NHANES) and simulated streams rather than live human implants.
- Proprietary Ecosystem & API Access: Market leaders in implantable biosensors (e.g., Abbott, Dexcom) utilize closed-source, proprietary ecosystems to protect intellectual property and adhere to strict FDA/CE regulations. These corporations do not grant third-party, learning-based research projects access to their raw data streams or private APIs. Consequently, this project is constrained to use Simulated IoT Ingestion, where the database is populated with synthetic telemetry that mimics the exact packet structure, frequency, and noise profiles of real-world sensors.

LITERATURE REVIEW

This chapter provides a critical analysis of the current state of diabetes management technology and clinical decision-support systems (CDSS). It establishes the theoretical foundation for the project and identifies the technological gaps that the proposed Agentic AI and Hybrid Database approach aim to bridge.

Introduction

The management of Type 2 Diabetes (T2D) is a lifelong challenge requiring continuous monitoring of glycemic levels and cardiovascular markers. Traditionally, this has been a reactive process, where treatments are adjusted only after laboratory tests reveal poor control. However, the rise of Digital Health and the Internet of Things (IoT) has shifted the focus toward proactive, real-time intervention. This review explores the evolution from manual monitoring to AI-driven systems, highlighting how modern architectures like TimescaleDB and Large Language Models (LLMs) are transforming patient outcomes.

Existing Solutions and Research Works

Current research in Diabetes Management Technology is divided into three distinct generations. Each generation solved a specific problem but introduced new limitations that this project addresses.

First Generation: Discrete Monitoring

The earliest digital solutions relied on Self-Monitoring of Blood Glucose (SMBG). These devices were simple but lacked data persistence.

Limitation: Data was often handwritten by patients, leading to "Recall Bias" and lost information.

Second Generation: Connected CGM Ecosystems

The introduction of Continuous Glucose Monitors (CGM) like Dexcom and Medtronic allowed for real-time tracking.

Limitation: As discussed in Chapter 1, these are "Proprietary Silos." The data is locked within the manufacturer's app, making it difficult for a doctor to correlate glucose with other factors like Blood Pressure or sleep.

Third Generation: AI-Driven CDSS

Recent academic works (2023–2025) have integrated Machine Learning to predict hypoglycemia.

Research Gap: Most current AI models are "Black Boxes." They might predict a risk but cannot explain the medical reasoning to the doctor. Furthermore, they lack a natural language interface for the patient.

Proposed Project vs. Existing Solutions

To demonstrate the technical necessity of this project, it is essential to compare it against the current market leaders in diabetes management. While applications such as mySugr, Glucose Buddy, and Dexcom Clarity provide excellent tracking interfaces, they function primarily as "Passive Data Loggers." Our proposed platform introduces an Agentic AI layer that shifts the paradigm from passive logging to active, clinical intervention.

Comparative Analysis: Market Leaders

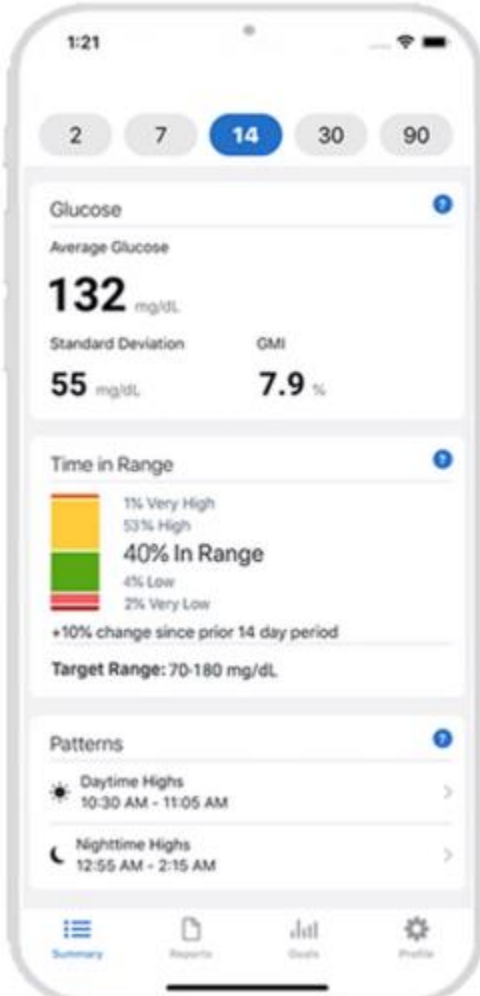
- **mySugr (Roche):** Highly effective for manual logging and "gamified" engagement. However, its Clinical Decision Support (CDSS) is limited to bolus calculators for insulin users and does not offer a doctor-validated medication titration engine for Type 2 patients.
- **Glucose Buddy:** Offers a comprehensive all-in-one tracker for glucose and blood pressure but relies on static "Educational Plans." It lacks a Generative AI assistant to answer patient-specific questions about lab results (e.g., ACR or Lipid profiles) in real-time.
- **Dexcom Clarity:** While it provides advanced retrospective data analysis and trend reports for users of Dexcom CGM systems, it is restricted to a "closed ecosystem." It lacks the flexibility to integrate multiple comorbidity markers (like chronic kidney disease data) into a single Time-Series Dashboard optimized for the high-velocity, multi-variable IoT telemetry that is a core feature of our TimescaleDB implementation.

The Research Gap: The "Intelligent Mediator"

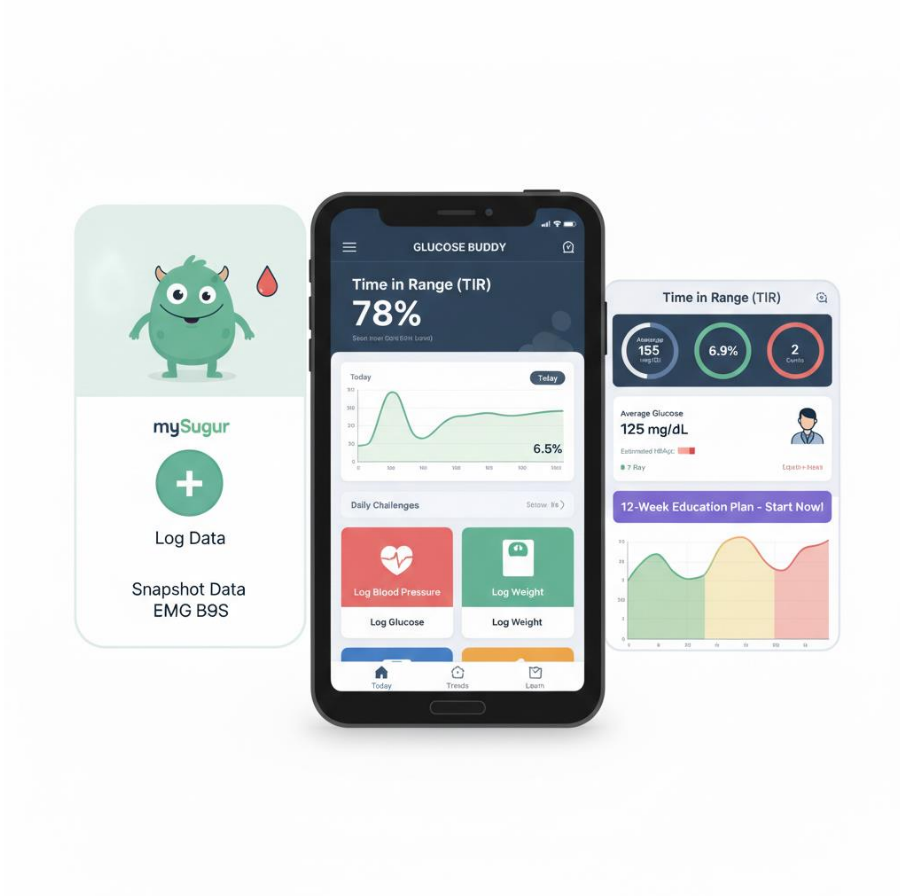
The primary gap in existing solutions is the lack of a "Doctor-in-the-Loop" Validation Logic. Most apps either provide no advice or provide automated alerts that the doctor has not reviewed. Our project solves this by ensuring that every clinical suggestion generated by the AI agent is stored in a MySQL suggestions table, awaiting a manual "Accept/Reject" from his doctor.

- Dexcom Clarity:

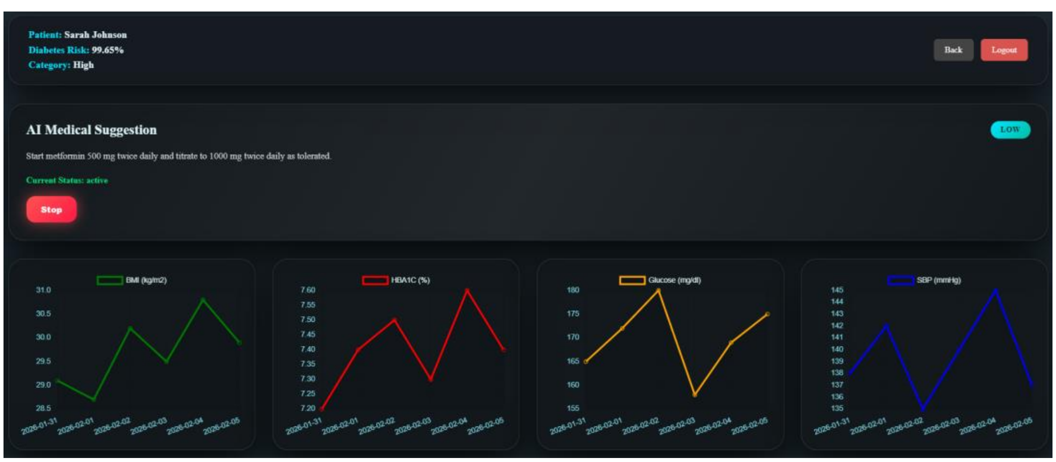
1Dexcom Clarity App Home Page

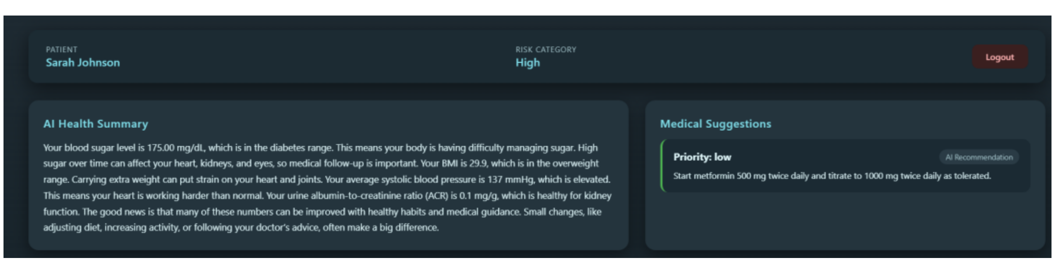


- Glucose Buddy:



- Agentic AI System for Type 2 Diabetes Monitoring and Management:





Comparison Table

2Comparison Table

Feature	mySugr	Glucose Buddy	Dexcom Clarity	Proposed Platform
Monitored Markers	Glucose Only	Glucose & BP	Glucose Only	Glucose, BP, HbA1c, BMI, Lifestyle (Sleep/Activity)
Data Engine	Standard SQL	Standard SQL	Proprietary Cloud	Hybrid (MySQL + TimescaleDB)
Intelligence Layer	Passive (Manual)	Static Plans (educational)	Predictive Alerts	Agentic AI (Llama-3 + CDSS + Predictive Modeling)
Prediction Scope	None	None	Glucose Trends	Multi-variable Risk Stratification
Decision Support	Bolus Calculator	None	Trend Alerts	ADA-Logic Medication Titration
Patient Interaction	Manual Logbook	Task Tiles	Hardware Sync	Real-time Conversational Assistant
Doctor Validation	No (PDF only)	No (PDF only)	Retrospective Review	Real-time Validator & Reviewer

Table 1Comparison Table Between Present Solutions

System Architecture

Introduction

The architecture of the proposed Agentic AI system for Type 2 Diabetes management is engineered to transition from passive patient monitoring to active, intelligent clinical intervention. Traditional healthcare systems often operate with delayed feedback loops, limiting the ability to provide real-time risk assessment and medication adjustments. To overcome these limitations, the system employs a modular, multi-layered design that integrates IoT-based sensing, predictive modeling, rule-based decision support, generative AI, and a Doctor-in-the-Loop verification mechanism.

This chapter details the technical blueprint of the platform, explaining the rationale behind each component, the flow of data through the system, and the architectural decisions that ensure scalability, safety, and clinical relevance.

The high-level system architecture of the proposed Agentic AI platform illustrates a streamlined, end-to-end flow of patient data from initial acquisition to visualization. This architecture emphasizes the following sequence:

- IoT Data Acquisition: The process begins with IoT Devices (wearables and medical sensors) capturing real-time biometric telemetry.
- Edge Transmission: Data is passed through an Edge Transmitter, which serves as a gateway to represent edge computing capabilities before sending information to the cloud.
- Hybrid Storage: Data is stored in a Hybrid Database (SQL + Postgre), providing a robust foundation for both structured clinical records and flexible time streams.
- Backend Processing: A FastAPI Backend (Python-based) manages the ingestion and routing of data, ensuring high-performance communication between the edge and storage layers.
- Agentic AI Analysis: The Agentic AI layer get data from the hybrid database to perform complex analysis, utilizing predictive and Rule based models to derive health insights.
- Actionable Insights: Finally, these insights are delivered to the Doctor Patient info Dashboard, where charts and graphs provide the doctor with a clear, actionable overview of Patient health status.

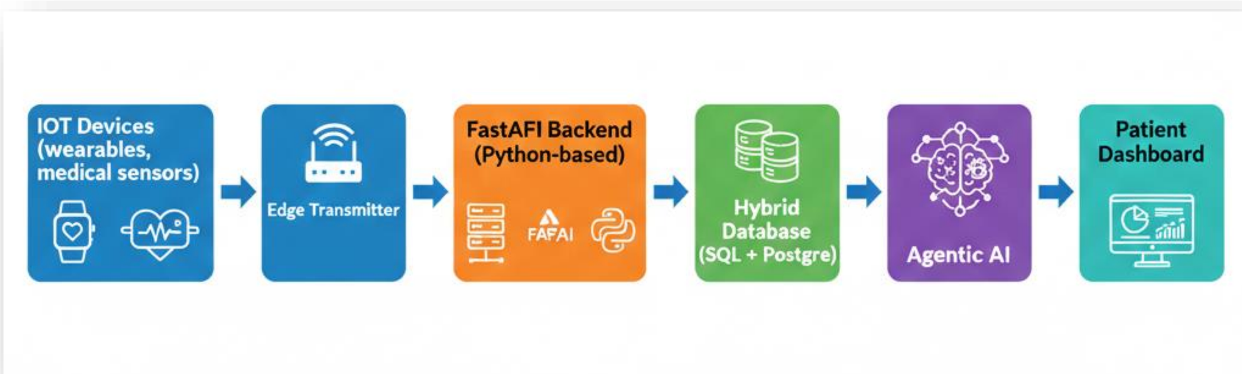


Figure 3High-level system architecture showing the technical flow from IoT devices through the edge transmitter, FastAPI backend, hybrid database storage, Agentic AI processing, Doctor-in-the-Loop validation, and final delivery to the patient dashboard.

System Overview

The platform is designed around five key functional layers:

- Sensing Layer (IoT Devices): Continuous Glucose Monitoring (CGM) sensors, blood pressure monitors, and lifestyle tracking devices, body mass index and Hba1c tracking.
- Edge Processing Layer: On-body or near-device processing unit that validates sensor signals, converts raw measurements to meaningful units, and transmits data to the backend.
- Backend API Layer: FastAPI handles asynchronous data ingestion, validation, database storage, and triggers predictive and generative AI modules, and Rule Based model.
- Intelligence Layer: Includes:
  - Predictive Modeling: Logistic Regression classifier for risk stratification
  - Rule-Based Logic (CDSS): Clinical guidelines-driven medication suggestions
  - Generative AI (LLM) for Diabetic Question Answering
- Human-in-the-Loop Layer: Doctor reviews all AI-generated suggestions, approves/rejects/stops actions, and provides final intervention.

This layered design allows the system to maintain high performance and low latency while ensuring safety-critical decisions remain under clinical supervision.

Hybrid Database Architecture

A central feature of the system is its hybrid database design, combining MySQL for relational data and TimescaleDB (PostgreSQL extension) for high-frequency time-series data.

Relational Layer (MySQL)

MySQL stores structured, non-temporal data:

Table 2 Table Showing Mysql Stuctural Tables

Table	Description
Doctors	Credentials, hashed passwords, verification status, license number

Patients	Demographics, doctor assignment, registration metadata
Devices	IoT hardware inventory, serial numbers, models, manufacturer
Suggestions	AI or doctor-generated clinical recommendations

Time-Series Layer (TimescaleDB)

TimescaleDB stores temporal clinical data, including:

- Measurements: glucose, HbA1c, SBP,
- Medications: history of drug administration and dosage change, and adherence
- Lifestyle logs: activity, sleep
- Predictions: historical outputs of the risk engine

Hybrid Strategy Justification

The hybrid design ensures strong performance across both transactional and analytical workloads.

MySQL was selected for structured relational data due to its mature tooling for credential management, while TimescaleDB optimizes high-frequency IoT telemetry.

The relational layer (MySQL) manages: sensitive patient and doctor records with structured schemas, complex joins, and reliable data validation.

The time-series layer (TimescaleDB) is tailored for high-frequency sensor streams, enabling efficient temporal queries and long-term risk trend analysis.

Together, these layers balance consistency with scalability, supporting both secure record-keeping and real-time health monitoring.

Database Architecture

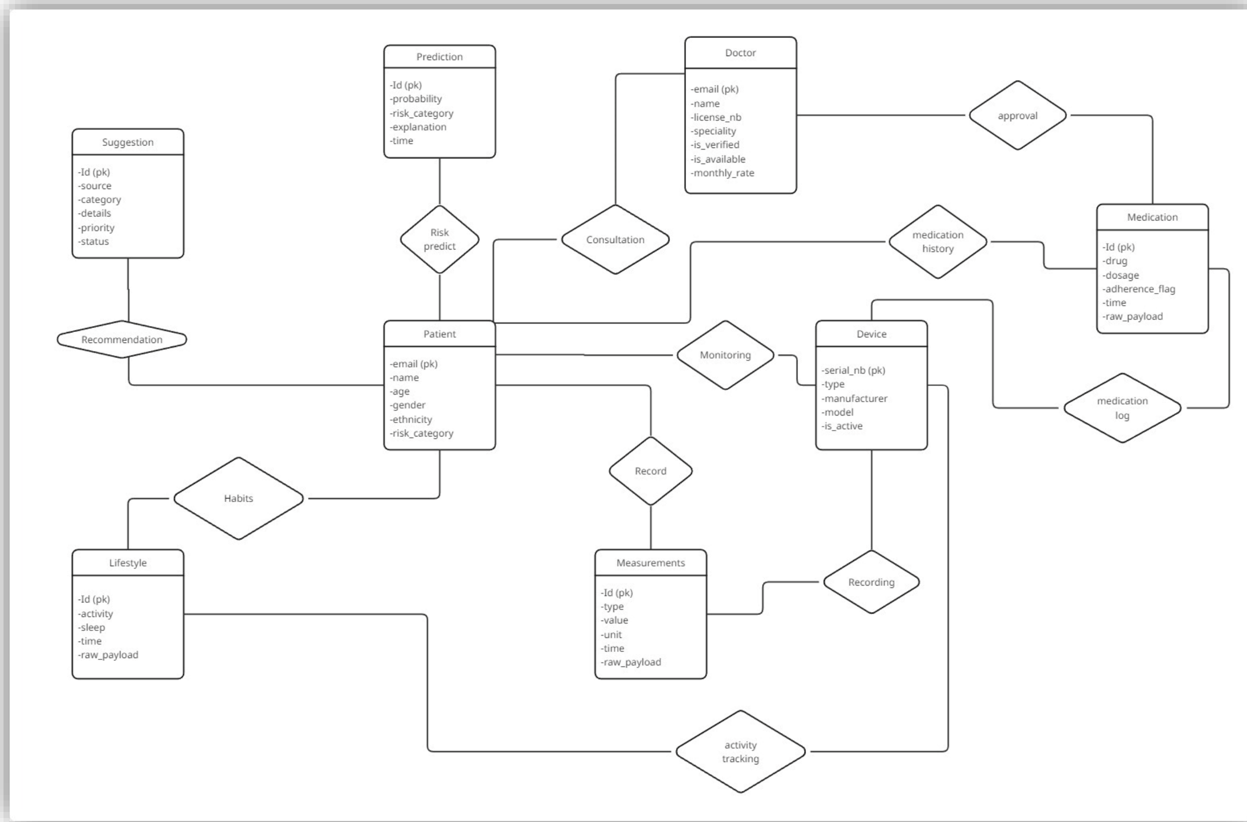


Figure 4Database Er Diagram

Data Flow and Agentic AI Loop

The system implements a closed-loop Agentic AI framework.

Step 1: Sensor Data Collection:

- IoT sensors send glucose, BP, and lifestyle data, Hba1c, BMI every 5 minutes.

Step 2: Edge Processing:

- On-device validation and unit normalization
- Basic anomaly detection to discard outliers

Step 3: API Ingestion:

- FastAPI endpoint validates incoming JSON payloads
- Stores data in TimescaleDB
-
```python
@router.post("/update_availability")        You, 12 hours ago • Second commit
async def update_doctor_availability(
    email: str = Form(...),
    status: bool = Form(...),
    mysql: AsyncSession = Depends(get_mysql_session)
):
    repo = MySQLDoctorsRepo(mysql)
    await repo.availability(email, status)
    return RedirectResponse(url=f"/dashboard", status_code=HTTP_303_SEE_OTHER)
```

-
```python
async with mysql_session_factory() as session:
    repo = MySQLSuggestionsRepo(session)
    medical_services = MedicalService(repo)
    suggestions = await medical_services.generate_suggestions(
        patient_email=patient["email"],
        hba1c=measurement_data_last_7_days.get('HBA1C', {}).get('avg_value', 0),
        glucose=measurement_data_last_7_days.get('Glucose', {}).get('avg_value', 0),        You, 12 hours ago • Second
        bmi=measurement_data_last_7_days.get('BMI', {}).get('avg_value', 0),
        sbp=measurement_data_last_7_days.get('SBP', {}).get('avg_value', 0)
    )
    suggestions = await repo.get_all_for_doctor(patient['email'])
    if not suggestions:
        suggestions = [{'details': '{"text": "There are no suggestions at the moment. Please check back later."}'}]
    for s in suggestions:
        if isinstance(s["details"], str):
            try:
                s["details"] = json.loads(s["details"])
            except Exception:
                pass
return templates.TemplateResponse("patient_info.html", {"request": request, "user": user,
                                                        "patient_measurement_data": measurement_data,
                                                        "prediction_data": prediction,
                                                        "sbp_data": sbp_data,
                                                        "bmi_data": bmi_data,
                                                        "hba1c_data": hba1c_data,
                                                        "glucose_data": glucose_data,
                                                        "patient": patient,
                                                        "suggestions": suggestions})
```

Figure 5Code Snip From Patient Info Dashboard Route

Figure 6Code Snipest for Api Routes

Step 4: Predictive Risk Stratification

- Logistic Regression model computes probabilities
- Categorize Probability into Low/Medium/High risk
- Explain The elevated or low risk

Step 5: Rule-Based Clinical Decision Support System (CDSS)

- Applies evidence based clinical rules from multiple organizations (ADA, CDC, WHO, NICE, Diabetes Canada, AACE, EASD, KDIGO, ESC, IDF, AHA/ACC, ACOG).
- Covers thresholds for HbA1c, fasting glucose, BMI, comorbidities, and patient risk factors.
- Generates structured intervention suggestions (e.g., lifestyle modification, metformin initiation, dual/triple therapy, insulin initiation, adjunctive treatments).
- Suggestions are stored in MySQL as "Suggestion" records with a proposed status for doctor review.
-

```python
# Glucose rule
if glucose >= 126 and hba1c < 6.5:
    suggestions.append(make_suggestion(
        "Start metformin 500 mg twice daily for impaired fasting glucose.",
        "low"
    ))

# BMI rule
if bmi >= 30:
    suggestions.append(make_suggestion(
        "Add semaglutide 0.25 mg weekly, titrate to 1 mg weekly for weight reduction.",
        "medium"
    ))

# SBP rule
if sbp >= 140:
    suggestions.append(make_suggestion(
        "Add lisinopril 10 mg daily or losartan 50 mg daily for hypertension control.",
        "high"
    ))

if suggestions:
    await self.suggestions.add_many(patient_email, suggestions)
return suggestions
```

Figure 7Code Snip for Rule Based Model Function

Step 6: Generative AI (LLM)

- Uses a finetuned Llama 3.2 1B model (via Unsloth).
- Dedicated to answering patient questions specifically related to diabetes.
- Provides natural language responses that are medically aligned and patient friendly.
- Focuses on education and explanation, not clinical decision making.

- Integrated into the patient dashboard to improve accessibility and understanding.
-
```
model, tokenizer = FastLanguageModel.from_pretrained(
    model_name="unsloth/Llama-3.2-1B-Instruct",
    max_seq_length=1024,
    dtype="float16",
    load_in_4bit=True
)

model = FastLanguageModel.get_peft_model(
    model,
    r=8,
    target_modules=["q_proj","k_proj","v_proj","o_proj","gate_proj","up_proj","down_proj"],
    lora_alpha=16,
    lora_dropout=0,
    bias="none",
    use_gradient_checkpointing="unsloth",
    random_state=3407,
)

print("✅ Base model ready for LoRA fine-tuning.")
```

Figure 8Code Snip For Loading Base Model

Step 7: Doctor-in-the-Loop Validation

- Physician reviews AI-generated recommendations
- Approves, modifies, or rejects intervention

Risk Stratification Engine

The predictive engine at the heart of the Agentic AI system is implemented using Logistic Regression, chosen for its interpretability, robustness, and ability to provide probabilistic outputs suitable for risk stratification. Logistic Regression enables the system to compute the probability that a patient is in a high-risk state given a set of physiological and lifestyle features.

The Model is Formally represented as:

- $$P(Y=1) = \frac{1}{1+e^{-(\beta_0+\beta_1 X_1+\cdots+\beta_n X_n)}}$$
- Where:
    - Y=1 as example represent High risk Diabetes.

- Xn are predictive features such as hba1c and glucose.
- Beta are the learned model coefficient representing the influence of each feature on the probability of being high risk

Calibration and Risk Categorization

To ensure the probabilities correspond to clinically meaningful risk levels, the model is calibrated using Platt Scaling via Scikit-learn's CalibratedClassifierCV. This calibration ensures risk scores approximate true probabilities, as validated, The calibrated output P(Y=1) is then mapped into three risk categories:

Low Risk: $P<0.35$

Medium Risk: $0.35 \leq P \leq 0.7$

High Risk: $P>0.7$

This mapping allows the system to trigger appropriate clinical responses and generate recommendations through the CDSS while avoiding false alarms.

```
final_model = Pipeline([
    ('preprocessing', preprocessor),
    ('model', LogisticRegression(max_iter=1000, class_weight='balanced'))
])
final_model.fit(X_train, y_train)
calibrated_model = CalibratedClassifierCV(estimator=final_model, method="sigmoid", cv=5)
calibrated_model.fit(X_train, y_train)
```

Figure 9Code Snip For Model Training and calibrating

Explain ability and Feature Interpretation

A key requirement of this project is interpretable AI — clinicians and patients must understand why a certain risk level is assigned. Logistic Regression inherently supports feature-level explain ability, as each coefficient Beta i reflects the direction and magnitude of the feature's influence on risk:

- Positive beta i increases the probability of high risk (e.g., elevated HbA1c).
- Negative beta i decreases the probability (e.g., high physical activity reducing risk).

The system generates human-readable explanations by combining:

- Coefficient interpretation: "HbA1c contributes significantly to high risk due to values above 8%."
- Threshold reasoning: "BMI > 30 contributes moderately to risk, increasing probability by 12%."
- Temporal context: By analyzing trends in time-series data, the model can identify sudden spikes or gradual deterioration, providing context-aware narratives such as: "Glucose levels increased consistently over the past 48 hours, indicating elevated hyperglycemia risk despite normal BMI."

Integration with the Agentic AI Loop

The Risk Stratification Engine does not operate in isolation; it is embedded in the three-tier Agentic AI framework:

- Rule-Based Layer: Uses thresholds to trigger immediate safety alerts.
- Predictive Layer (Logistic Regression): Assigns probabilistic risk and identifies contributing features with explanation.
- Generative Layer (LLM): Answer patient Diabetic Question.

This integration ensures that every patient risk assessment is interpretable, actionable, and clinically safe, reinforcing the Doctor-in-the-Loop validation before any therapeutic suggestion reaches the patient.

Security Considerations

Authentication & Authorization:

- Passwords hashed with argon2id/BCrypt via PassLib CryptContext
- Role-based access control for doctors and patients

Data Privacy:

- TLS/HTTPS for all API endpoints
- Encrypted WebSocket channels for real-time chat

Database Security:

- Strict access control policies
- Isolation between relational and time-series data layers

Summary

- This Chapter has presented a comprehensive design blueprint for the Agentic AI system:
- Hybrid database design balances relational integrity and high-frequency telemetry performance
- Closed-loop Agentic AI architecture ensures proactive, interpretable risk assessment
- CDSS and Doctor-in-the-Loop framework maintain clinical safety while leveraging AI efficiency
- Layered modularity supports scalability, maintainability, and extensibility

User Architecture

Introduction

The User Architecture of the Agentic AI system defines how patients, doctors, and administrators interact with the platform. While Chapter 4 focused on the technical backbone, this chapter explains the user-centric view, illustrating the system's usability, workflow, and interfaces. The architecture ensures that all users can access actionable insights in a secure, intuitive, and clinically meaningful way.

The platform implements a role-based, multi-layer interface design:

- Patients: Receive real-time risk notifications, explanations of their health metrics, and education through the Agentic AI conversational interface.
- Doctors: Review AI-generated clinical suggestions, monitor patient health trends, and communicate with patients through a secure dashboard in addition for medical adherence check.
- Administrators: Manage Doctor verification

The user architecture ensures that complex AI decisions are presented clearly, maintaining safety, transparency, and engagement.

User Roles and Responsibilities

Table 3Different Users and their role

Role	Responsibilities	Interface Components
Patient	Monitor personal health, view risk scores, receive education, get suggestion	web dashboard, LLM Page to chat with ai.
Doctor	Validate AI suggestions, review patient history, manage treatment adjustments	Web dashboard, risk visualization charts, suggestion review panels
Admin	Manage Doctor validations	Admin dashboard

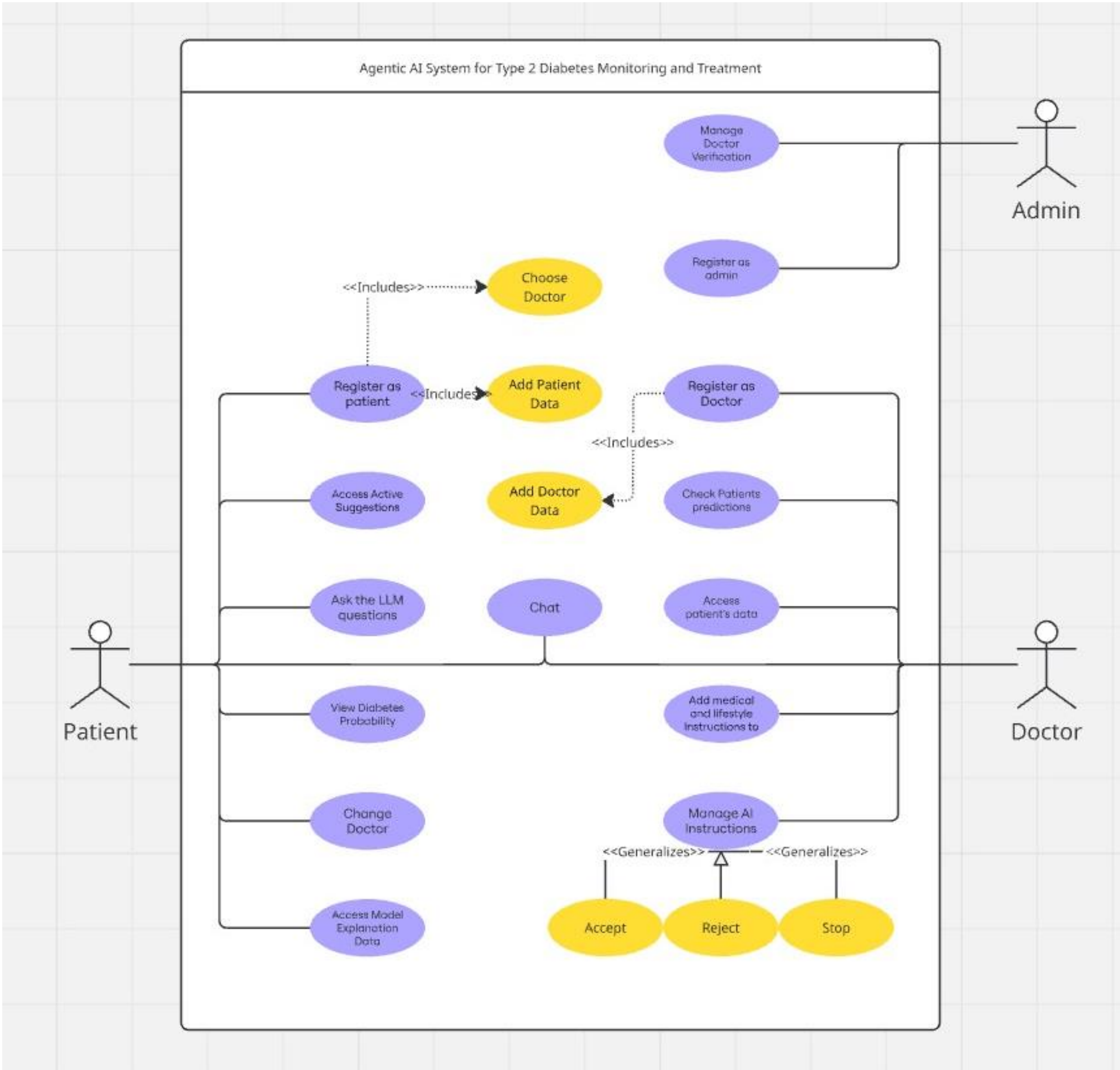


Figure 10User Case Diagram

User Interaction Layers

The user experience is designed in four main layers:

Presentation Layer

- Provides a responsive interface for web and mobile users.
- Displays risk categories, health trends, AI recommendations, and notifications.

- Visualizations include time-series graphs (glucose, HbA1c, BP), trend indicators, and interactive dashboards.

Interaction Layer

- Supports bidirectional communication between users and the system.
- Patients can ask questions through the Agentic AI chat.
- Doctors can accept, reject, or modify suggestions with a single click.
- Real-time updates are pushed via WebSockets for instantaneous feedback.

Logic Layer

- Maps user actions to backend processes:
  - Risk assessments → updated in patient dashboards
  - AI suggestions → staged for doctor review
  - Patient queries → routed to LLM for explanation

Ensures role-based access control, so each user sees only the relevant data**.**

Security & Compliance Layer

- Handles authentication, authorization, and encryption.
- Ensures patient data privacy under TLS/HTTPS channels and encrypted WebSocket connections.

Patient Workflow

Step 1: Onboarding

- Patients register and input personal data (age, gender, ethnicity…).
- Devices are linked and synchronized with the system.

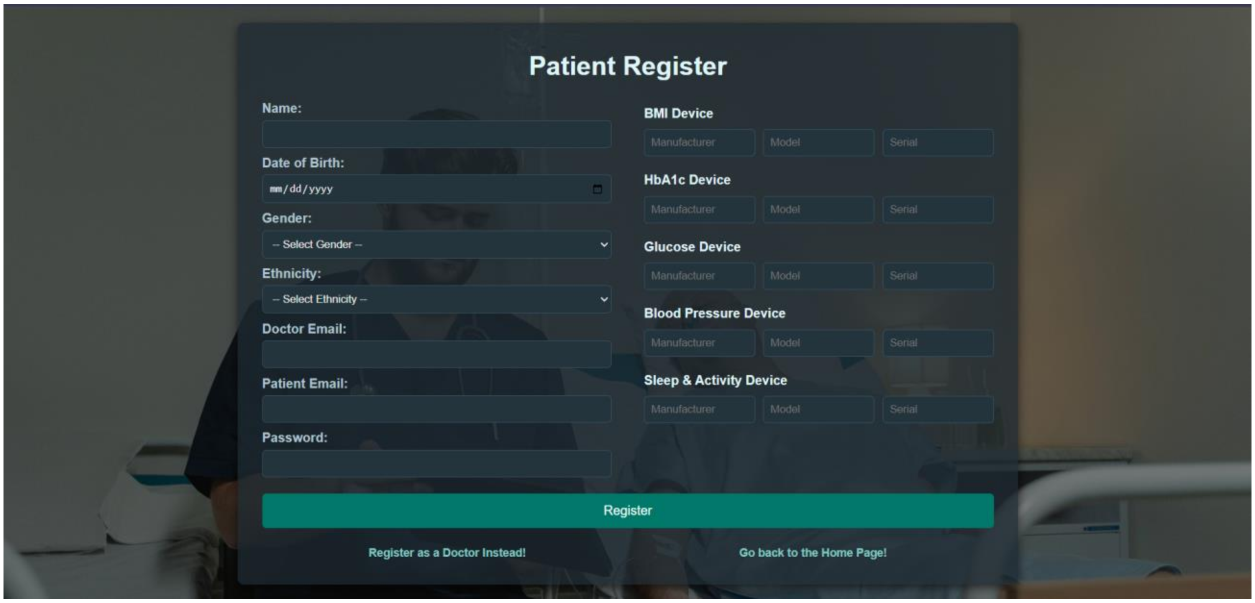
- 

Figure 11Patient Register Page

Step 2: Data Collection

- IoT sensors continuously feed biometric data.
- Edge devices perform initial validation and push data to the database.

Step 3: Risk Awareness

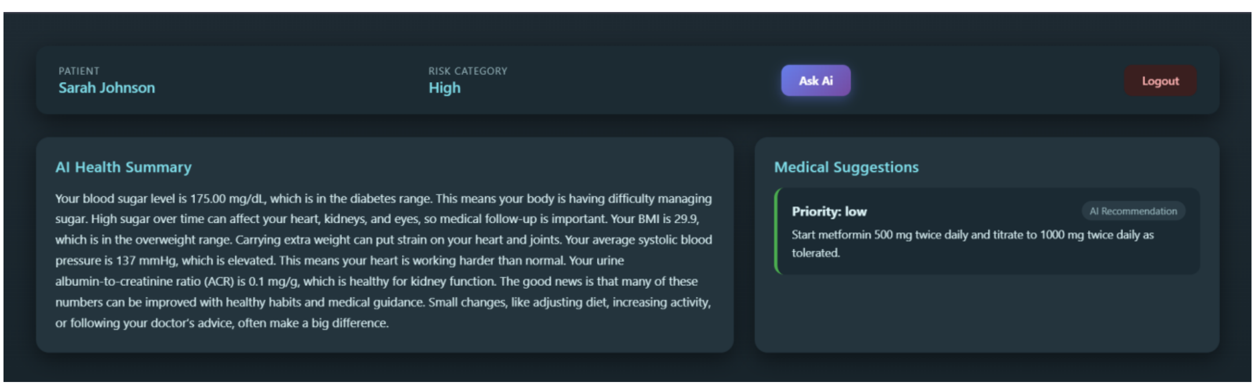
- Patient dashboard displays current risk status (Low, Medium, High) using color-coded visualizations.
- Risk explanations generated by the prediction model using coefficient rule-based explanation provide contextual understanding of trends.
- Active Suggestion provided by doctor or ai
- 

Figure 12Patient Dashboard Page

Step 4: Education & Guidance

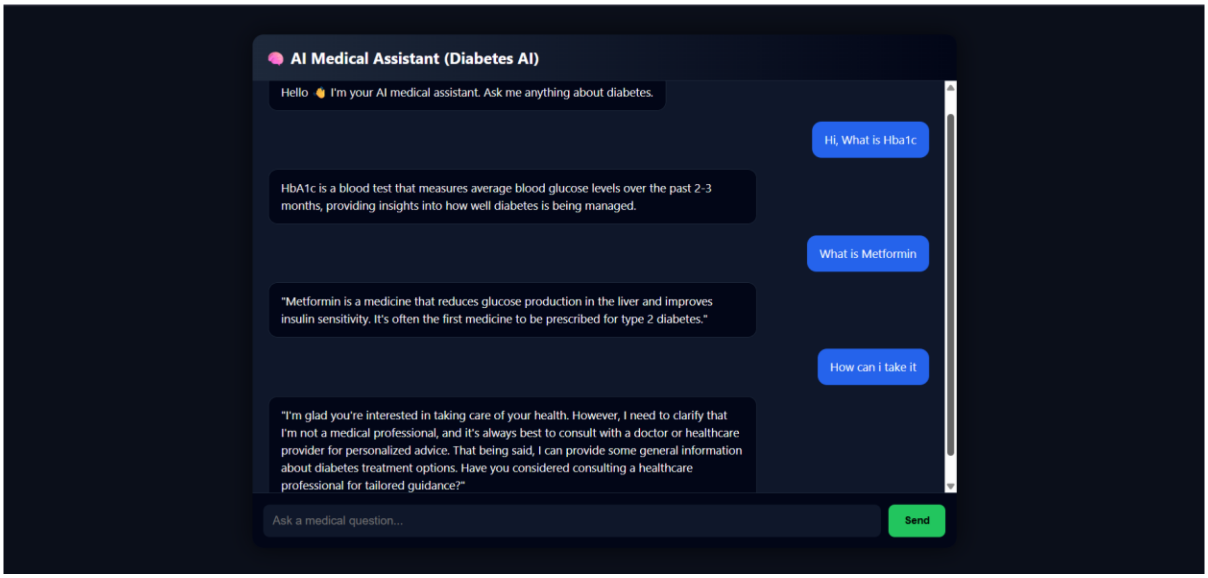
- Large Language Model answers patient questions:
    - "They suggested to me to take metformin, can you explain What is metformin?"
    - "How does sleep affect my HbA1c?"
- Explanations are medically accurate and tailored to the patient's metrics.
- 

Figure 13LLM Chating page

Step 5: Notifications & Alerts

- Immediate alerts are sent if critical thresholds are exceeded, with recommendations for next steps.
- All alerts maintain a Doctor-in-the-Loop validation, ensuring safety.

Doctor Workflow

Registration, Credential Verification, and System Onboarding

Before accessing any patient data, doctors must undergo a formal onboarding and verification process:

- Doctors register using professional credentials (license number, specialization).

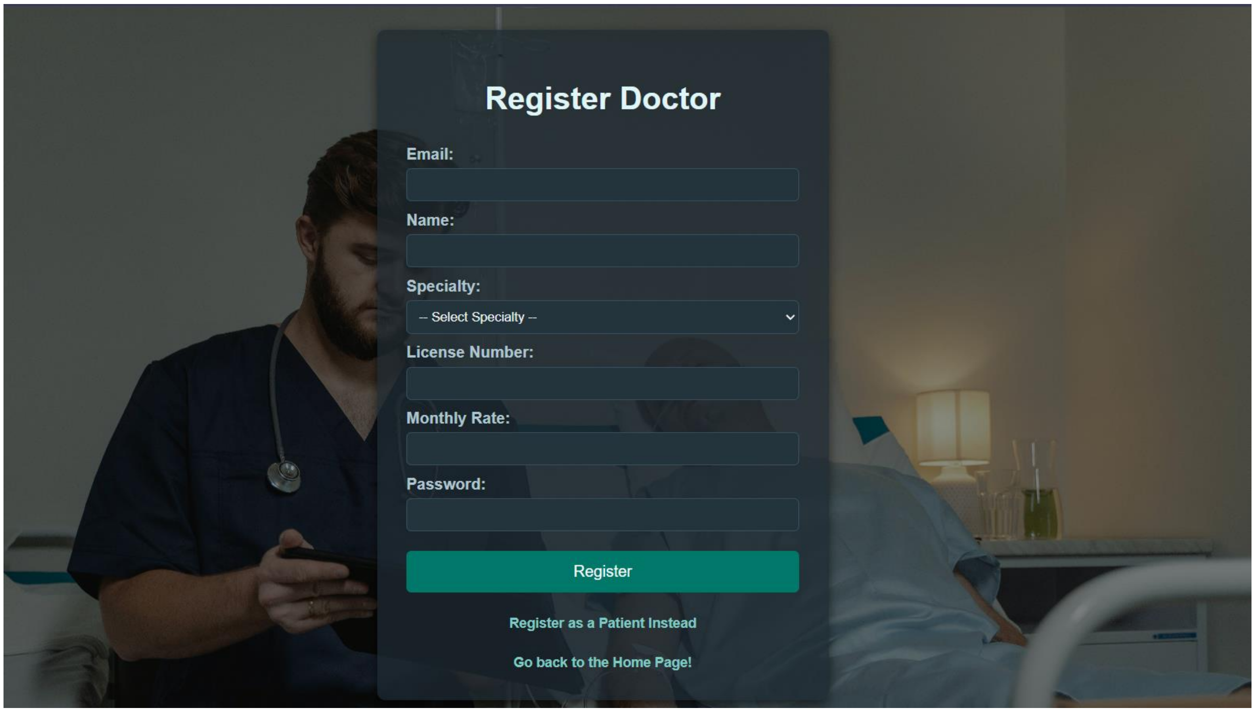
- 

Figure 14Doctor Register Page

- Submitted credentials are stored in the Database with a pending verification status.
- An administrator verifies the doctor's identity and license authenticity.
- Only verified doctors can assign patient and benefit from the website.

This process ensures legal compliance, data security, and clinical accountability.

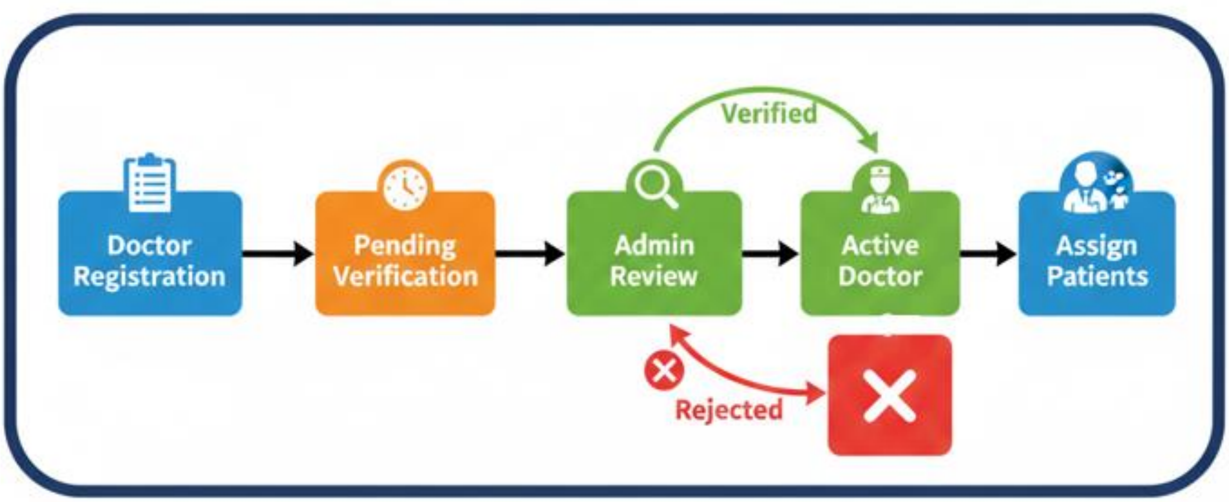


Figure 15Doctor registration and admin verification workflow diagram

Patient Monitoring

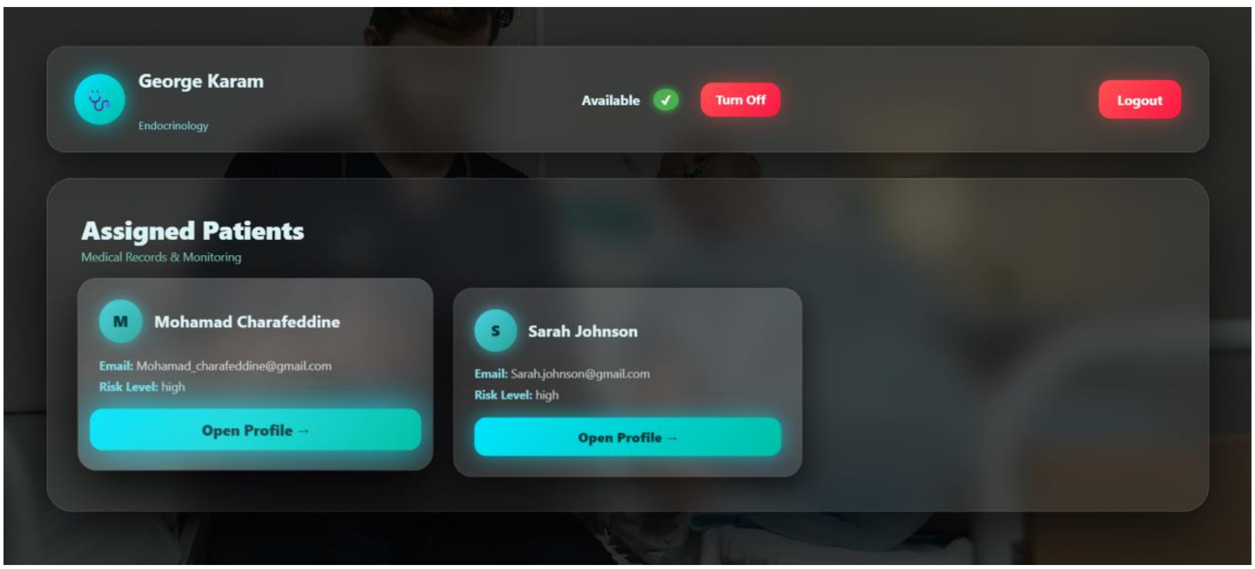
- Verified doctors access real-time dashboards showing risk trends across all assigned patients.
- Time-series visualizations support detection of long-term patterns and short-term anomalies.
- The system provides clinicians with integrated treatment adherence monitoring, enabling:
  - Analysis of medication compliance
  - Missed or delayed doses
  - Long-term adherence trends
  -
- This allows doctors to clearly distinguish between Physiological disease progression and Behavioral non-adherence ensuring accurate clinical interpretation and decision-making.
- Risk categorization (Low/Medium/High) allows prioritization of high-risk patients.
- Medication adherence tracking
- 

Figure 16Doctor Dashboard

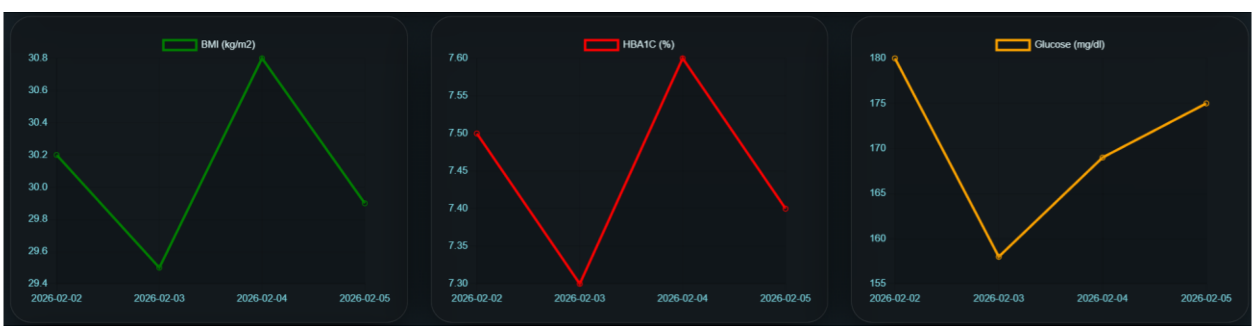
- 

Figure 17Time Series Visualization for Patient Data In Doctor Patient Information Page

Suggestion Review

AI-Agent suggestions appear as structured clinical items within the doctor dashboard interface, including:

- Medication adjustments
- Lifestyle recommendations
- Risk mitigation strategies

Each suggestion includes:

- Proposed intervention (e.g., medication change, lifestyle adjustment, treatment escalation)
- Current state (proposed, approved, rejected)
- Priority level (Low / Medium / High urgency classification)

Validation & Feedback

Doctors are provided with explicit control actions, allowing them to:

- Activate (approve) the recommendation
- Reject the recommendation
- Stop an active Treatment plan

This ensures that no AI-generated intervention is executed without human clinical validation, enforcing the Doctor-in-the-Loop safety model and supporting transparent, explainable, and accountable AI-assisted decision making.

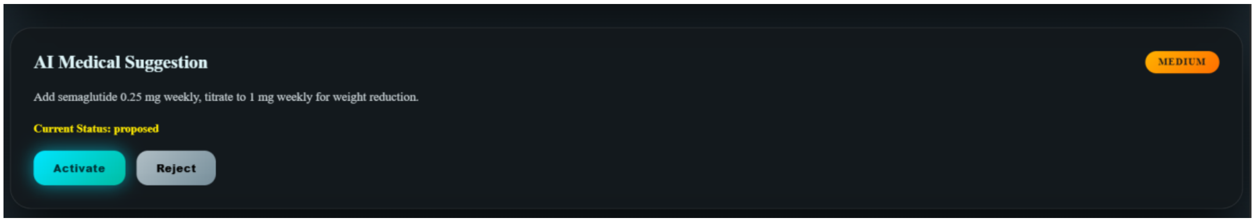


Figure 18Suggestion Waiting to Change status in Doctor Page

Clinical Communication

- Secure messaging enables doctors to explain decisions directly to patients.
- Encrypted chat channels ensure confidentiality and compliance.

- All interactions are logged for traceability, auditing, and continuity of care.

Admin Workflow

The Admin role represents the system-level governance authority responsible for platform integrity, professional verification, and regulatory control. Unlike doctors and patients, the Admin does not participate in clinical decision-making or medical interactions. Instead, the Admin acts as a trust and validation layer that ensures the legitimacy, safety, and institutional reliability of the healthcare platform.

The Admin responsibilities are limited to three core functions:

Doctor Verification and Accreditation

The admin is responsible for validating doctor accounts before they are authorized to access the clinical system. This includes:

- Reviewing submitted professional credentials
- Verifying medical licenses and certifications
- Approving or rejecting doctor registration requests

Only verified doctors are granted access to patient dashboards, AI suggestions, and clinical decision interfaces and even can Assign Patients ensuring that all medical interactions occur under licensed professional supervision.

Report Handling and System Oversight

The admin manages system-level reports related to doctors, including:

- Reviewing complaints or reports submitted about doctor behavior
- Investigating misuse of the system
- Applying administrative actions when required (warnings, suspension, access revocation)

Overall User Interaction Model

The overall user interaction model encapsulates how patients, doctors, and administrators engage with the Agentic AI system, forming a closed-loop ecosystem that ensures real-time monitoring, interpretability, and clinical safety. By integrating continuous IoT data streams, predictive risk stratification, AI-generated recommendations, and Doctor-in-the-Loop validation, the system provides an end-to-end workflow that balances automation with human oversight.

At the core of this model, patients act as both data providers and recipients of educational insights. IoT devices continuously capture physiological and lifestyle data, which is processed at the edge and transmitted securely to the backend. Patients access their dashboards to view real-time risk levels, trends, and AI-generated explanations. The LLM conversational agent enhances understanding by delivering context-aware medical guidance, improving patient engagement, literacy, and agency in managing Type 2 Diabetes. And The Rule Based Model provided and accepted by doctor suggestions will be shown in this dashboard

Doctors serve as the clinical authority within the loop. They monitor patient data, review AI-generated intervention suggestions, and provide final approval before any action is implemented. This ensures that all recommendations comply with established clinical guidelines and maintains accountability and safety. The system prioritizes high-risk patients, allowing physicians to allocate attention efficiently and intervene proactively.

Administrators provide system governance, managing doctor verification, system integrity, and compliance oversight. By controlling access and auditing interactions, they ensure that the platform remains secure, legally compliant, and reliable for all stakeholders.

The interaction model highlights several key principles:

- Proactive Monitoring: Continuous data collection enables timely identification of risks and anomalies.
- Interpretability: Risk stratification and AI suggestions are presented with clear, understandable explanations for both patients and doctors.
- Role-based Access: Each user interacts with the system according to their responsibilities, maintaining privacy and operational integrity.
- Closed-Loop Feedback: AI predictions, rule-based suggestions, and human validation form an iterative loop that continuously refines patient care.
- Actionable Insights: The system converts raw data into clinically meaningful guidance, bridging the gap between patient-generated data and professional medical decisions.

In conclusion, the Overall User Interaction Model demonstrates how the Agentic AI system integrates technology, intelligence, and human oversight into a cohesive framework. It ensures that patients are empowered with knowledge, doctors are supported in clinical decision-making, and administrators maintain secure and compliant operation. This structured interaction enables real-time, data-driven, and safe management of Type 2 Diabetes, transforming conventional healthcare into a proactive, intelligent, and patient-centered ecosystem.

ECONOMICAL, ETHIC, SAFETY AND SOCIAL IMPACTS

Introduction

In the development of innovative healthcare technologies, such as an Agentic AI system for Type 2 Diabetes management, assessing the broader impacts beyond technical performance is crucial. Projects that involve medical data, patient interactions, and AI-assisted decision-making can have far-reaching consequences on the economy, ethics, safety, and social structures.

The scope of this chapter is to provide a comprehensive evaluation of the economic, ethical, safety, and social impacts of deploying this system. By systematically analyzing these dimensions, the chapter aims to highlight potential benefits, identify challenges, and propose considerations for responsible implementation. This assessment ensures that technological innovation aligns with societal values, patient well-being, and sustainable development objectives.

Economic Impacts

The deployment of an Agentic AI system for Type 2 Diabetes management carries significant economic implications across multiple dimensions:

Cost-Benefit Analysis

Implementing this AI-driven healthcare platform requires upfront investment in hardware (IoT sensors, edge devices), software development (backend, AI models, dashboards), and ongoing maintenance. However, these costs are offset by potential long-term benefits:

- Reduced Hospitalizations: Early detection of high-risk events may decrease emergency visits and hospitalization rates.

- Improved Treatment Efficiency: Automated clinical suggestions reduce the time doctors spend analyzing patient data.
- Preventive Care Savings: By proactively managing diabetes, the system can lower costs associated with long-term complications like nephropathy or cardiovascular disease.

Job Creation

While automation may reduce certain administrative tasks, the system creates new roles, including:

- Data Analysts and AI Specialists: To maintain, update, and fine-tune AI models.
- Healthcare IT Support: For system deployment, troubleshooting, and integration.
- Healthcare Educators: To assist patients in understanding AI-generated insights.

Overall, the platform shifts job roles from routine data management to higher-value, specialized positions.

Local Economy

Deploying the system may stimulate local businesses in technology and healthcare sectors:

- Local IT companies may provide cloud hosting, device integration, and maintenance services.
- Healthcare providers could adopt new business models, including subscription-based telemedicine services.

Long-Term Sustainability

Economic viability over time depends on continuous updates, integration of new devices, and scalability:

- Software Scalability: The FastAPI backend and hybrid database design support growth without proportional cost increase.
- Hardware Upgrades: Modular sensor and edge device design reduces replacement costs.
- Healthcare Savings: Long-term reduction in diabetes-related complications may offset ongoing operational costs, making the system sustainable and economically attractive for both providers and patients.

Ethical Impact

The implementation of an Agentic AI system for Type 2 Diabetes management raises important ethical considerations that must be carefully addressed to ensure responsible and fair use.

Moral Considerations

- Patient Autonomy: Patients must retain control over their treatment decisions, even when AI provides clinical suggestions.
- Informed Consent: Users should be fully informed about how their data is collected, analyzed, and used to generate AI-driven recommendations.
- Bias and Fairness: AI models must be trained on diverse datasets to avoid biased risk predictions that could disadvantage certain patient groups.

Stakeholder Perspectives

- Patients: Expect transparency, clear explanations, and safety assurances when interacting with AI-generated health insights.
- Healthcare Providers: Require accurate, reliable recommendations and the ability to override AI decisions when necessary.
- Marginalized Groups: Special attention is needed to ensure equitable access, culturally sensitive communication, and inclusion in training data to prevent systemic biases.

Corporate Responsibility

- Organizations deploying such systems are responsible for maintaining privacy, accuracy, and accountability.
- Ethical responsibility extends to:
  - Regular audits of AI models to identify potential errors or biases.
  - Clear reporting mechanisms for patients or clinicians to flag issues.
  - Ensuring AI recommendations never replace professional clinical judgment, maintaining a Doctor-in-the-Loop framework as a safeguard.

By addressing these ethical considerations, the system aligns with principles of fairness, transparency, and patient-centered care, supporting trust in AI-assisted healthcare.

Project-Specific Implementation

In this project, several measures were implemented to address these ethical considerations:

- Doctor-in-the-Loop Framework: Every AI-generated suggestion is stored with a “proposed” status in MySQL and requires doctor approval before execution, ensuring patient safety and autonomy.
- Informed Consent: Patients register and are informed that their biometric data and AI-generated explanations are used for monitoring and education.
- Bias Mitigation: The Logistic Regression model was trained on the NHANES dataset with diverse demographics, and risk predictions are accompanied by interpretable explanations.
- Transparency & Accountability: The fine-tuned LLM provides natural language explanations for patient metrics, improving understanding, while all interactions are logged for auditing.

These implementations demonstrate how ethical considerations were integrated into the design and operation of the system, ensuring responsible, transparent, and patient-centered AI use.

Safety Impacts

Ensuring patient safety is a critical aspect of deploying an Agentic AI system for Type 2 Diabetes management. The platform’s design incorporates multiple layers of safeguards to minimize risks while supporting proactive healthcare.

Risk Assessment

- Clinical Risk Mitigation: The system categorizes patients into Low, Medium, or High-risk groups based on continuous physiological monitoring. Early identification of high-risk patients reduces the likelihood of severe complications.
- System Reliability: Edge processing validates sensor data and filters out anomalous readings before analysis, preventing false alarms or incorrect recommendations.
- AI Validation: All suggestions from the predictive engine or generative AI are reviewed by a certified physician before implementation, reducing the risk of unsafe autonomous actions.

Regulations and Standards

- The platform is designed in compliance with healthcare data protection standards such as HIPAA and GDPR, ensuring secure storage and transmission of patient data.

- Cryptographic measures, including TLS/HTTPS for APIs and encrypted WebSocket channels, provide secure real-time communication.
- Role-based access control ensures that only authorized doctors and patients can access sensitive information.

Impact on Public Safety

- By providing continuous monitoring and early alerts, the system can prevent acute diabetic events like hypoglycemia or hyperglycemia crises, enhancing overall patient safety.
- The Doctor-in-the-Loop framework ensures that no medication adjustment is executed without professional oversight, mitigating the risk of medical errors.
- The logging of all AI suggestions, doctor approvals, and patient interactions allows audit trails, supporting traceability and accountability in clinical decision-making.

Social Impacts

The deployment of an Agentic AI system for Type 2 Diabetes management has several social implications, affecting communities, public health, and access to healthcare resources.

Community Effects

- Enhanced Health Awareness: By providing patients with interpretable risk scores and explanations, the system encourages healthier lifestyles and greater engagement in personal health management.
- Empowering Caregivers: Family members and caregivers can better understand patient health trends through notifications and dashboards, improving support networks.
- Cultural Considerations: Patient communication via the LLM is designed to be clear and culturally sensitive, promoting trust and adoption across diverse populations.

Public Health

- Preventive Care: Continuous monitoring allows early intervention, reducing hospitalizations and long-term complications, which benefits overall community health.
- Chronic Disease Management: By proactively managing Type 2 Diabetes, the system contributes to lowering the societal burden of chronic diseases, including related cardiovascular and kidney complications.

- Health Literacy: The conversational AI educates patients on medical concepts (like HbA1c, BMI, or lifestyle impact), improving population-level understanding of diabetes

Equity and Access

- Inclusivity: The system's web-based interface and IoT integration ensure accessibility for patients with varying technical literacy, promoting equitable care.
- Bridging Gaps: Remote monitoring reduces geographic barriers to healthcare, allowing patients in underserved or rural areas to receive timely interventions.
- Digital Divide Considerations: While offering improved access, care is taken to ensure that patients without constant internet or device access are still able to benefit through periodic data uploads and offline notifications.

Project-Specific Social Measures

- The LLM's natural language explanations empower patients to understand their health, reducing anxiety and increasing confidence in self-management.
- The real-time doctor-patient communication fosters stronger clinical relationships and community trust in AI-assisted healthcare.
- By simulating implantable sensor telemetry and offering remote dashboards, the system demonstrates potential social benefits even in contexts where full-scale deployment is not yet feasible.

Conclusion

The assessment of economic, ethical, safety, and social impacts demonstrates that the proposed Agentic AI system for Type 2 Diabetes management is not only technically innovative but also socially responsible and sustainable. Economically, the platform has the potential to reduce healthcare costs through preventive monitoring while creating opportunities for skilled jobs in AI and healthcare technology. Ethically, the Doctor-in-the-Loop framework, informed consent, bias mitigation, and transparent AI explanations ensure patient autonomy, fairness, and accountability. Safety is reinforced through continuous monitoring, edge validation, and secure communication, minimizing clinical risks and supporting adherence to regulatory standards such as HIPAA and GDPR. Socially, the system empowers patients, caregivers, and communities by enhancing health literacy, enabling preventive care, and promoting equitable access to medical guidance regardless of location or demographic differences.

Collectively, these considerations illustrate that the Agentic AI platform not only advances technological capabilities but also aligns with societal values, ethical responsibilities, and sustainable healthcare objectives, reinforcing its relevance and potential for real-world adoption.

Project Management

Tasks and Schedule

Managing a complex project such as the development of an Agentic AI system for Type 2 Diabetes requires careful planning and scheduling of tasks to ensure timely and successful completion. This section outlines the project's key activities, their sequencing, and the schedule followed throughout the academic year.

The project was divided into ten main tasks, covering the full lifecycle from research and system design to implementation, testing, and documentation. Each task was assigned a specific start and end date, as well as a responsible individual (in this case, the project lead, Mohamad Ali Charafeddine).

The schedule was designed to:

- Prioritize core system components such as IoT data ingestion, predictive risk engine, and Rule-Based CDSS.
- Allow parallel development of complementary tasks like backend API development and LLM fine-tuning to optimize time efficiency.
- Include milestones and reviews to validate progress, refine designs, and ensure alignment with project objectives.
- Support iterative development, allowing feedback from supervisors and testing results to guide the next phases.

The following task table (Section 6-2) provides a structured overview of the tasks, responsibilities, timelines, and completion statuses. A visual representation of the schedule is provided in the Gantt diagram (Section 6-3), illustrating task durations, overlaps, and dependencies.

This structured approach ensured that all phases of the project were organized, trackable, and executed efficiently, resulting in the successful development of a fully functional Agentic AI platform for diabetes management.

Task Table

Table 4Task Table

Task ID	Task Description	Responsible	Start Date	End Date	Status
T1	Literature Review and Requirement Analysis	Mohamad Ali Charafeddine	01/10/2025	15/10/2025	Completed
T2	System Architecture Design	Mohamad Ali Charafeddine	16/10/2025	31/10/2025	Completed
T3	Database Design (MySQL + TimescaleDB)	Mohamad Ali Charafeddine	01/11/2025	10/11/2025	Completed
T4	Backend Development (FastAPI)	Mohamad Ali Charafeddine	11/11/2025	30/11/2025	Completed
T5	Predictive Risk Engine (Logistic Regression)	Mohamad Ali Charafeddine	01/12/2025	10/12/2025	Completed
T6	Rule-Based CDSS Implementation	Mohamad Ali Charafeddine	11/12/2025	20/12/2025	Completed
T7	LLM Fine-Tuning (Llama 3.2-1B)	Mohamad Ali Charafeddine	21/12/2025	05/01/2026	Completed
T8	Frontend & Dashboard Development	Mohamad Ali Charafeddine	06/01/2026	20/01/2026	Completed
T9	Testing & Debugging (Full System)	Mohamad Ali Charafeddine	21/01/2026	05/02/2026	Completed
T10	Documentation & Report Writing	Mohamad Ali Charafeddine	06/02/2026	08/02/2026	Completed

Gantt Diagram

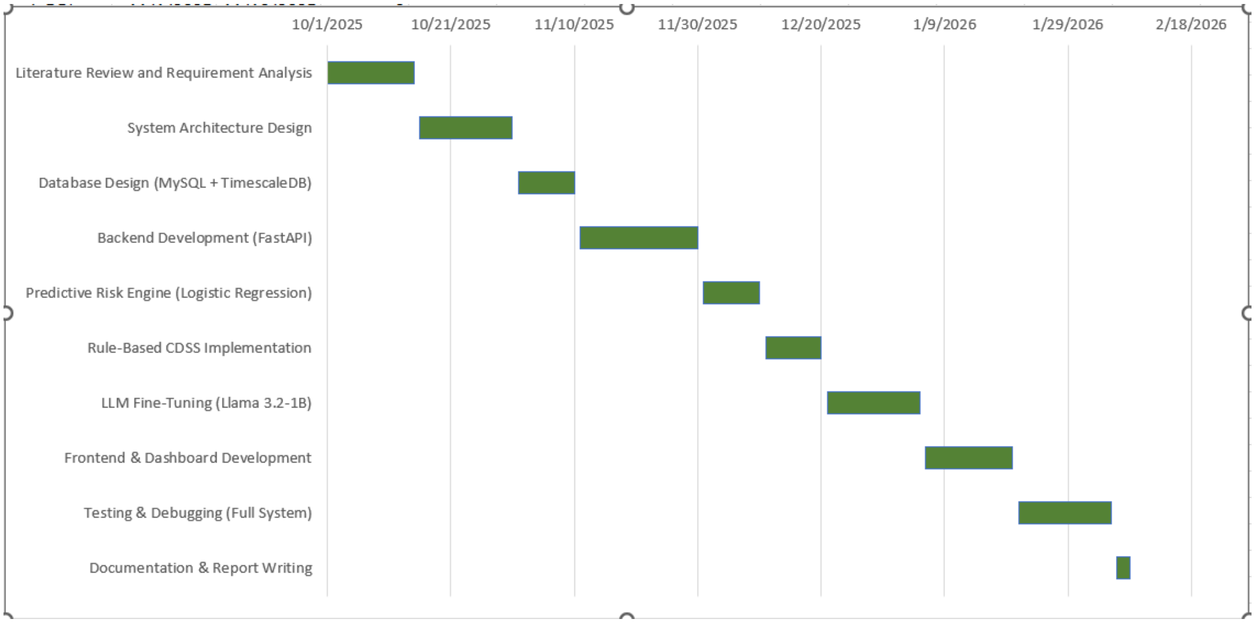


Figure 19Gantt Diagram

Conclusion and Future Work

Conclusion

This capstone project successfully developed an Agentic AI system for Type 2 Diabetes monitoring and management, integrating IoT-based real-time data collection, predictive risk stratification, rule-based clinical decision support, and a fine-tuned conversational AI for patient education.

The system demonstrates a closed-loop, Doctor-in-the-Loop architecture that ensures AI-generated recommendations are always validated by medical professionals, providing a safe, interpretable, and actionable healthcare solution. Key achievements include:

- Hybrid Database Design: Efficiently storing structured clinical data in MySQL and high-frequency time-series IoT data in PostgreSQL/TimescaleDB.
- Predictive Risk Engine: Logistic Regression model with calibrated probabilities, interpretable feature contributions, and actionable risk categories.

- Rule-Based CDSS: Evidence-based medication and lifestyle suggestions aligned with ADA, AACE, and WHO clinical guidelines.
- Agentic Conversational AI: Fine-tuned Llama-3.2-1B Instruct model providing patient-friendly explanations of lab results and diabetic knowledge.
- Secure Frontend Dashboards: Real-time visualization for both patients and doctors, with bidirectional communication and intervention tracking.

The project illustrates the potential of Agentic AI in chronic disease management, bridging the gap between continuous patient data collection and professional clinical oversight, while empowering patients with understandable health insights.

Future Work

While the project achieved its primary objectives, there are several directions for further development and improvement:

- IoT Device Integration: Currently, the system uses simulated IoT streams. Future work could focus on integrating real wearable or implantable devices, either by collaborating with medical professionals or leveraging publicly available APIs from commercial devices (e.g., Dexcom, Abbott, FreeStyle Libre). This will allow real-time, live physiological data to feed directly into the Agentic AI system.
- Multi-Morbidity Support: Expanding the platform to monitor and manage other chronic conditions, such as hypertension, cardiovascular disease, or kidney disease, using a unified predictive and rule-based framework.
- Enhanced LLM Capabilities: Fine-tuning the conversational AI further on multi-lingual datasets, patient behavior trends, or broader health literacy questions to improve accessibility and personalized patient education.
- Mobile Application Development: Creating a mobile app interface to complement the web dashboard, allowing patients to receive alerts, interact with the AI assistant, and communicate with their doctor on-the-go.
- Clinical Trials and Validation: Conducting pilot studies under supervision of licensed healthcare professionals to evaluate the platform’s effectiveness, usability, and clinical safety in real-world scenarios.

By pursuing these improvements, the system could evolve into a fully integrated, real-world healthcare solution, providing continuous, proactive, and personalized management for Type 2 Diabetes and potentially other chronic conditions.

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